习题课，第一章

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4. $P(A)=0.7$, $P(A-B)=0.3$ $\Rightarrow \begin{cases} P(A)=0.7 \\ P(A)-P(AB)=0.3 \end{cases}$

$\Rightarrow P(AB)=0.4$

$P(\overline{AB}) = 1-P(AB) = 1-0.4 = 0.6$

$(P(C) = 1-P(\bar{C}))$

5. 解: $P(A\cup B\cup C)$

$= P(A)+P(B)+P(C)-P(AB)-P(AC)-P(BC)+P(ABC)$

$0 \le P(ABC) \le P(AB) = 0 \Rightarrow P(ABC)=0$

$\therefore P(A\cup B\cup C) = 3/4 - 1/8 = 5/8$.

6. 古典概型: $P = \frac{|A|}{|\Omega|}$

$$P(A) = \frac{C_a^2 + C_b^2}{C_{a+b}^2}$$

$P(B) = 1 - P(A)$

条件概率:

$$P(A) = \frac{a}{a+b} \times \frac{a-1}{a+b-1} + \frac{b}{a+b} \times \frac{b-1}{a+b-1}$$

7. 解(1) 古典概型: $P = \frac{C_3^3}{C_{10}^3} = \frac{1}{120}.$

条件概率: $P = \frac{3}{10} \times \frac{2}{9} \times \frac{1}{8} = \frac{1}{120}$

(2) $P = (\frac{3}{10})^3$

9. 解：(1) $p=\dfrac{C_8^3}{C_{12}^3}$

(2) $p=\dfrac{C_8^2\cdot C_4^1}{C_{12}^3}$

(3) $p=1-\dfrac{C_8^3}{C_{12}^3}$

(4) $p=\dfrac{C_8^3+C_4^3}{C_{12}^3}$

10. 解：(1) $p=1-\dfrac{364^{500}}{365^{500}}$

(2) $p=\dfrac{C_6^4\times C_{12}^1\times 11^2}{12^6}$

15. 解：A_i：有 i 件次品，$i=0,1,2,3,4$

B：取10件产品检测出一件次品。

C：次品不超过两件。

$$P(C|B) = \underline{P(A_0|B) + P(A_1|B) + P(A_2|B)}$$

$$P(A_i|B) = \frac{P(A_i)\cdot P(B|A_i)}{P(B)}$$

$$P(B) = \sum_{i=0}^{4} P(A_i)\cdot P(B|A_i)$$ 草稿内容。

依题意，$P(A_0)=0.35$，$P(A_1)=0.25$

$P(A_2)=0.2$，$P(A_3)=0.18$，$P(A_4)=0.02$

$$P(B|A_0) = 0.$$

$$P(B|A_1) = \frac{C_{49}^{9} \cdot C_{1}^{1}}{C_{50}^{10}} = \frac{1}{5}$$

$$P(B|A_2) = \frac{C_{48}^{9} \cdot C_{2}^{1}}{C_{50}^{10}} = \frac{16}{49}$$

$$P(B|A_3) = \frac{C_{47}^{9} \cdot C_{3}^{1}}{C_{50}^{10}} = \frac{39}{98}$$

$$P(B|A_4) = \frac{C_{46}^{9} \cdot C_{4}^{1}}{C_{50}^{10}} = \frac{988}{2303}$$

全概率公式；

$$P(B) = \sum_{i=0}^{4} P(A_i) \cdot P(B|A_i)$$

$$= 0.196$$

$$P(A_0|B) = \frac{P(A_0)\cdot P(B|A_0)}{P(B)} = 0$$

$$P(A_1|B) = \frac{P(A_1)\cdot P(B|A_1)}{P(B)}$$

$$= \frac{0.25 \times \frac{1}{5}}{0.196} = 0.255$$

$$P(A_2|B) = \frac{P(A_2)\cdot P(B|A_2)}{P(B)} = 0.333.$$

$$P(C|B) = P(A_0|B) + P(A_1|B) + P(A_2|B)$$

$$= 0 + 0.255 + 0.333$$

$$= 0.588$$

16. 解： A_1：损坏2%， A_2：损坏10%

A_3：损坏90%.

B：取三件，三件均是好的.

目的： $P(A_i|B)$ ，$i=1,2,3$.

$$P(A_i|B)=\frac{P(A_iB)}{P(B)}.$$

$$P(B)=P(A_1)\cdot P(B|A_1)+P(A_2)\cdot P(B|A_2)+P(A_3)\cdot P(B|A_3).$$

依题意知： $P(A_1)=0.8$， $P(A_2)=0.15$，

$P(A_3)=0.05$

$$P(B|A_1)=0.98\times0.98\times0.98=0.98^3$$

$$P(B|A_2)=0.9^3$$

$$P(B|A_3)=0.1^3$$

$$P(B) = 0.8624.$$

$$P(A_1|B) = \frac{P(A_1)\cdot P(B|A_1)}{P(B)} ✓$$

$$P(A_2|B) = \frac{P(A_2)\cdot P(B|A_2)}{P(B)} ✓$$

$$P(A_3|B) = \frac{P(A_3)\cdot P(B|A_3)}{P(B)} ✓$$

$$P(A_1|B) > P(A_2|B) > P(A_3|B)$$

∴ 损坏率为2%.

17. 解：A_i：有 i 件次品，$i=0,1,2$

B：通过验收（取4件无次品）

(1) 求 $P(B)$.

$$P(B)=\sum_{i=0}^{2} P(A_i)\cdot P(B|A_i)$$

$P(A_0)=0.8$，$P(A_1)=0.15$，$P(A_2)=0.05$

$P(B|A_0)=1$

$$P(B|A_1)=\frac{C_{23}^4}{C_{24}^4}=5/6$$

$$P(B|A_2)=\frac{C_{22}^4}{C_{24}^4}=\frac{95}{138}.$$

$\Downarrow$

$P(B)=0.96$.

(2) 解：$\beta = P(A_0|B)$

$$= \frac{P(A_0)\cdot P(B|A_0)}{P(B)}$$

$$= \frac{0.8\times 1}{0.96} = \frac{1}{12}$$